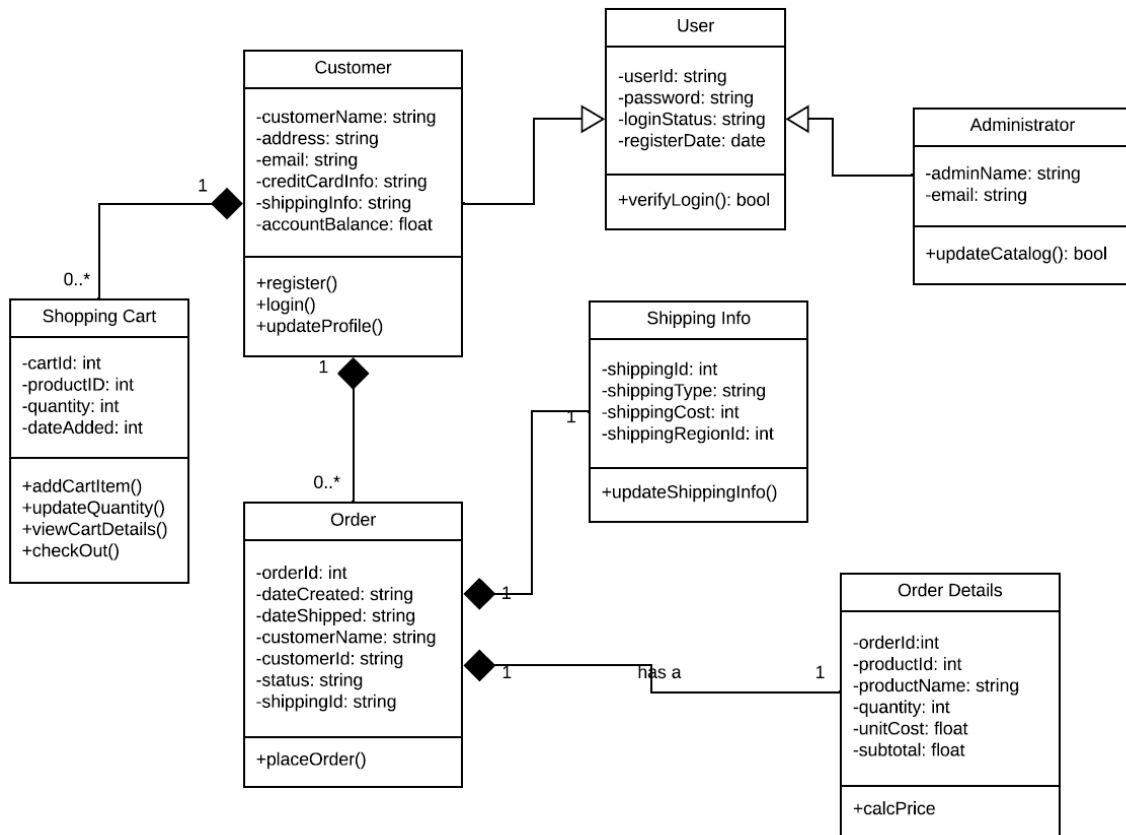


Sample 1

Online Shopping System Description

The **User** is the primary component of the Online Shopping System. Each user has a **userId**, **password**, **loginStatus**, and **registerDate**. Users are required to verify their login credentials before accessing the system. The **Admin** is responsible for controlling and managing the entire system. The admin can update the product catalog and manage customer information. Additionally, the admin's **name** and **email** are visible to customers. A **Customer** is a registered user who can browse products, add products to a shopping cart, and place orders. Each customer has a **customerId**, **name**, **address**, **email**, **creditCardInfo**, **balance**, and **shoppingDetails**. Customers can select products, manage their shopping carts, and complete the checkout process to place orders. The **Shopping Cart** stores the products selected by customers before purchase. It contains **cartId**, **productId**, **quantity**, and **dateAdded** as its attributes. The shopping cart provides functionalities such as **Update Cart**, **Checkout**, and **View Cart Details**, allowing customers to modify their selected products and complete their purchases. When a customer confirms a purchase, an **Order** is created. Each order contains **orderId**, **orderDate**, **dateOfShipment**, **customerName**, **customerId**, **orderStatus**, and **shippingId**. During the order process, the system asks the customer to choose a **payment method** before the order is finalized. Each order consists of one or more **Order Details**, which include **orderId**, **productId**, **productName**, **quantity**, and **totalPrice**. These details record the products purchased in the order along with their quantities and total prices. Finally, the **Shipping** component manages the delivery information for each order. It contains **shippingId**, **shippingType**, and **shippingCost** as its attributes and provides the **Update Shipping Details** operation to ensure accurate shipment information. Together, these components interact to provide a complete online shopping system where users can securely log in, customers can purchase products, admins can manage the system, and orders are processed and shipped efficiently.

Solution



Sample 2

Central Banking System Description

The **Central Bank** is the highest authority in the banking system and controls every other bank within a country. It has **name**, **location**, and **code** as its attributes. The Central Bank is the only entity authorized to **control reserves**, and it also provides loans to other banks to maintain financial stability. A **Bank** operates under the supervision of the Central Bank. Each bank has a **name**, **location**, and **code**. It is responsible for maintaining its branches, managing customer accounts, and monitoring its resource balance. Banks ensure the smooth operation of financial services through their branches. A **Branch** represents a local office of a bank. Each branch contains **cityInformation** and **branchId** as its attributes. It provides operations such as **Add Account**, **Maintain Account**, and **Remove Account**, enabling customers to open, manage, or close their bank accounts. An **Account** is owned by a customer and contains **accountUserName**, **accountNumber**, and **accountCredentials**. It provides various banking services, including **Take Loan**, **Add Money**, **Withdraw Money**, and **Manage Loan**, allowing customers to perform everyday financial transactions and manage their loans. A **Savings Account** is a specialized type of account that includes all the features of a standard account while also providing **incentives to customers**, such as interest on deposited money, encouraging them to save regularly. A **Student Account** is designed specifically for students. It helps students maintain their account balance by offering banking services suited to their financial needs and often includes special benefits or reduced service charges. A **Corporate Account** is intended for business organizations and corporate clients. It enables companies to manage their corporate financial affairs efficiently and securely, supporting large-scale business transactions and financial operations. The **Customer** is the end user of the banking system. Customers can open different types of accounts, perform financial transactions, deposit and withdraw money, apply for loans, and manage their banking activities through their respective bank branches. Together, the Central Bank, banks, branches, accounts, and customers form a complete banking system that supports secure and efficient financial management.

Solution

